

5.3 Retrieval, Bias, and Robustness

Nearest neighbors in two directions, and the stereotypes that come with them.

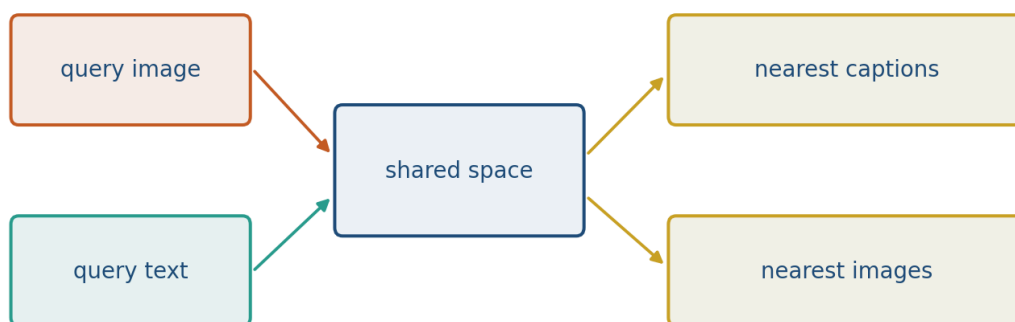
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Once image and text share a space, **retrieval** is nearest neighbors. The same geometry that makes search work also encodes **stereotypes** from the training web.

1. Two directions

Text → **image**: a query string, rank photos. **Image** → **text**: a photo, rank captions or documents. Dual retrieval is why coordinated towers beat a joint vector that you can only query from one side.

One space, two retrieval directions

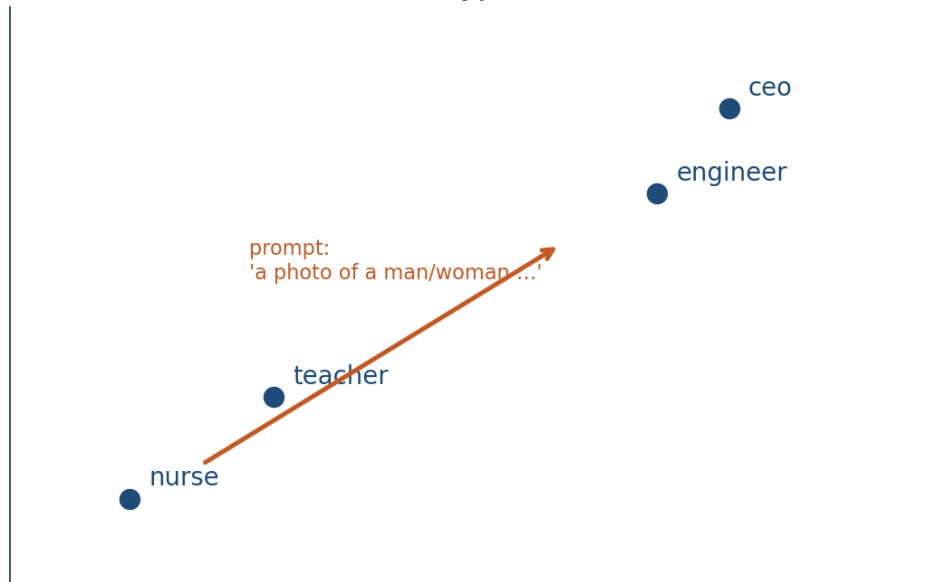


Report **recall@k** and a few failure cases (wrong object, wrong count, text that ignores the image). For a project, freeze the gallery embeddings; do not re-encode millions of images on every demo.

2. Bias and robustness

Occupation and gender, geography and skin tone, "a terrorist" versus a news photo: CLIP-style models pick up the co-occurrences of the web. Zero-shot prompts (a photo of a man who is a nurse) move the query along the same directions you saw in Week 1's toy embeddings.

VLMs inherit dataset stereotypes in retrieval and zero-shot



Robustness: typographic attacks (text painted in the image), distribution shift (medical, sketches), and language variety. A high ImageNet zero-shot number does not certify your domain.

If your project ranks people, jobs, or health content, put a probe in the report: a small, documented set of prompts and the retrievals they return. Debiasing is incomplete; measuring is required.

3. Practice

1. You can retrieve with a caption but not with a photo. What did you probably throw away?
2. Why is recall@1 a poor single number for a 100k image gallery?
3. Write one probe prompt you would be embarrassed to ship, and what you would count as a bad retrieval.